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Theoretical Physics

Quantum Field Theory II

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Introduction

In these notes, we will explore the fascinating world of Quantum Field Theory (QFT). QFT is a fundamental framework in physics that combines classical field theory, special relativity, and quantum mechanics. It provides a powerful language for describing the behavior of particles and fields at the most fundamental level. We assume are going to assume that the reader has a basic understanding of quantum mechanics and special relativity, as well as some familiarity with classical field theory and quantum field theory of free fields (Which will anyway be reviewed in the first chapter).

There exist different motivations for studying QFT: it provides a framework for describing the interactions between particles and fields, and it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. Additionally:

- QFT is the language of the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity); it is used for the quantization of systems with an **infinite number of degrees of freedom**, such as fields; it is essential for understanding the behavior of particles at high energies and small distances, where quantum effects become significant; and it provides a framework for describing the behavior of particles in curved spacetime, which is important for understanding phenomena such as black holes and the early universe.
- It is a successful framework for describing the behavior of particles and fields at the most fundamental level, incorporating both quantum mechanics and special relativity:
 - At high energies we can witness **particles creation and annihilation**, which cannot be described by non-relativistic quantum mechanics; QFT provides a framework for describing these processes, in an infinite-dimensional system, where the number of particles is not fixed.
 - QFT is a relativistic theory, which means that it is consistent with the principles of special relativity, implmenting **causality** and **Lorentz invariance**; this is essential for describing the behavior of particles at high energies and small distances, where relativistic effects become significant. So two events that are spacelike separated cannot influence each other

$$(x - y)^2 = (x^0 - y^0)^2 - \sum_{i=1}^3 (x^i - y^i)^2 < 0 \implies [\mathcal{O}(x), \mathcal{O}(y)] = 0,$$

thus the events are causally disconnected, ensuring causality.

- Particles are described as excitations of underlying fields, which can be created and annihilated; this allows for a more complete description of particle interactions and the behavior of particles at high energies. Examples of particles and underlying fields include:
 - **Scalar field:** $\phi(x) \implies$ spin-0 particles (e.g., Higgs boson).
 - **Spinor field:** $\psi(x) \implies$ spin-1/2 particles (e.g., electrons, quarks).
 - **Vector field:** $A_\mu(x) \implies$ spin-1 particles (e.g., photons, W and Z bosons).
- The **Standard Model** of particle physics is a quantum field theory that describes the fundamental particles and their interactions (except for gravity); it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. It is a quantum field theory that describes the electromagnetic, weak, and strong interactions, through the exchange of gauge bosons among fields of spin 0, 1/2, and 1 particles.
- The inclusion of gravity in a quantum field theory framework is an open problem in theoretical physics, but can be approached by inserting the metric tensor of general relativity

$$g^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu}.$$

While there are approaches to quantizing gravity, such as string theory and loop quantum gravity, a complete and consistent theory of quantum gravity has not yet been developed. Quanta of the gravitational field are called gravitons, which are hypothetical spin-2 particles that mediate the gravitational interaction, but they have not been observed experimentally. These notes will not cover quantum gravity, but it is an important area of research in theoretical physics which sometimes will be mentioned throughout the notes.

The main goal of these notes is to provide a comprehensive understanding of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level, as well as mathematical methods for making theoretical predictions on interactions among particles. We will cover a wide range of topics, including:

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- The quantization of free fields, which provides the foundation for understanding the behavior of particles and fields in quantum field theory.
- The construction of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level.
- The use of perturbation theory and Feynman diagrams to calculate scattering amplitudes and other physical observables in quantum field theory.
- The renormalization of quantum field theories, which is necessary to make sense of the infinities that arise in perturbation theory and to extract meaningful physical predictions from the theory.
- The application of quantum field theory to the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity).

QFT is one of the most successful and powerful frameworks in physics, and it has been instrumental in our understanding of the fundamental nature of the universe. This theory has been tested and

confirmed by a wide range of experiments, some of which tested the predictions of QFT to an extraordinary degree of precision. For example, the anomalous magnetic moment of the electron has been measured to an accuracy of one part in a trillion ($\sim 10^{-13}$), and the predictions of QFT have been confirmed to this level of precision. Additionally, the discovery of the Higgs boson at the Large Hadron Collider in 2012 provided further confirmation of the predictions of QFT and the Standard Model.

Another essential aspect of these notes is to provide a comprehensive understanding of the importance of symmetries in quantum field theory, which play a crucial role in determining the behavior of particles and fields. We will explore the concept of gauge symmetries, which are local symmetries that arise in quantum field theories and are responsible for the interactions between particles and fields. We will also discuss the role of global symmetries, which are symmetries that do not depend on spacetime coordinates, and their implications for the behavior of particles and fields. In the end we will argue that *a QFT is essentially defined by its fields and its symmetries* (statement that will be made more precise in the end of the notes), and we will see how symmetries can be used to constrain the behavior of particles and fields, and to make predictions about their interactions.

For a more comprehensive introduction to quantum field theory, we recommend the following textbooks:

- **An Introduction to Quantum Field Theory** by Michael E. Peskin and Daniel V. Schroeder: This is a widely used textbook that provides a comprehensive introduction to quantum field theory, covering both the theoretical foundations and practical applications of the subject.
- **Quantum Field Theory** by Mark Srednicki: This textbook offers a clear and concise introduction to quantum field theory, with an emphasis on the physical intuition behind the mathematical formalism.
- **Quantum Field Theory in a Nutshell** by A. Zee: This book provides a more informal and intuitive introduction to quantum field theory, making it accessible to a wider audience while still covering essential topics.
- **The Quantum Theory of Fields** by Steven Weinberg: This three-volume series is a comprehensive and rigorous treatment of quantum field theory, written by one of the pioneers in the field. It is suitable for advanced students and researchers who want a deep understanding of the subject.

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For more information about the notation and conventions used in these notes, please refer to the appendix, where we provide a detailed explanation of the symbols and mathematical conventions that we will use throughout the notes (Appendix A).

Prerequisites: *Familiarity with the Lagrangian and Hamiltonian formalisms of Classical Mechanics, standard Quantum Mechanics, Special Relativity and the basics of Quantum Field Theory (free fields) is required.*

1 | Free QFT Review

A free quantum field theory does not account for interactions between particles. It describes particles that do not interact with each other, and the fields that represent them evolve independently. Free QFTs are often used as a starting point for understanding more complex interacting theories, as they provide a simpler framework to study the properties of quantum fields and particles.

The lagrangian for a free quantum field theory typically consists of kinetic terms and mass terms for the fields, without any interaction terms: it can only contain **quadratic terms** in the fields, without any higher-order interactions. Quanta of the free fields correspond to particles that do not interact with each other, and their dynamics are governed by the **free propagators** of the theory, which describe how particles propagate through spacetime without any interactions: propagators are the Green's functions of the free theory, and they determine how particles propagate from one point to another in spacetime with determined probability amplitudes.

We will review the free quantum field theories for different types of particles, including scalar fields (spin 0), fermionic fields (spin 1/2), and gauge fields (spin 1). We will also discuss the symmetries of these theories and how they lead to conservation laws through Noether's theorem. This review will provide a foundation for understanding more complex interacting quantum field theories, and it will also highlight the differences between free and interacting theories, as well as the role of symmetries in quantum field theory.

1.1 | Spin 0

For a real scalar field $\phi(x)$, the free Lagrangian is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2, \quad (1.1.1)$$

where m is the mass of the scalar particle. The equation of motion derived from this Lagrangian is the **Klein-Gordon equation**:

$$(\square + m^2)\phi = 0, \quad (1.1.2)$$

where $\square = \partial_\mu \partial^\mu$ is the d'Alembertian operator. The solutions to this equation describe free scalar particles with mass m . This is a classical equation of motion, having the same role of Maxwell's equations in classical electromagnetism, but for a scalar field $\phi(x)$ instead of a vector field $A^\mu(x)$.

The idea of quantum field theory is to promote the classical field $\phi(x)$ to a quantum operator $\hat{\phi}(x)$ that acts on a Hilbert space of states. The quantization procedure involves imposing commutation relations on the field operators, which leads to the creation and annihilation of particles. The free scalar field can be expanded in terms of creation and annihilation operators, which allows us to describe the particle content of the theory. Indeed, we can see how decomposing the field into its Fourier modes leads to the interpretation of the quanta of the field as particles with specific momenta and energies:

$$\phi(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{\phi}(t, \mathbf{k}),$$

where each mode can be computed by using this ansatz in the Klein-Gordon equation in momentum space, leading to the dispersion relation of an harmonic oscillator represented by on-shell particles with energy $E_{\mathbf{k}} = \omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$:

$$(\partial_t^2 + \omega_{\mathbf{k}}^2) \tilde{\phi}(t, \mathbf{k}) = 0,$$

showing how the free scalar field can be interpreted as a collection of independent harmonic oscillators, each corresponding to a different momentum mode.

In order to quantize the free scalar field, we have to promote the fields themselves to operators and impose the canonical commutation relations, which leads to the creation and annihilation of particles. Before showing how to do that, it is crucial to note that $\hat{\phi}(x)$ depends on both space and time, enforcing the Heisenberg picture of quantum mechanics, where operators evolve in time while states remain fixed. The canonical commutation relations for the scalar field are given by

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}),$$

where $\hat{\pi}(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \hat{\phi}(x))} = \partial_0 \hat{\phi}(x)$ is the conjugate momentum operator. This constraint along with reality of the field $\hat{\phi}(x) = \hat{\phi}^\dagger(x)$ allows us to express the field operator in terms of **creation and annihilation operators**:

$$\hat{\phi}(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-ik\cdot x} + \hat{a}_{\mathbf{k}}^\dagger e^{ik\cdot x} \right),$$

where $k \cdot x = \omega_{\mathbf{k}} t - \mathbf{k} \cdot \mathbf{x} = k^\mu x_\mu$, and $\omega_{\mathbf{k}} = k^0$. The creation and annihilation (ladder) operators satisfy the following commutation relations

$$\begin{cases} [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}'), \\ [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] = [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] = 0. \end{cases}$$

Note that while the field operator $\hat{\phi}(x)$ is Hermitian and time dependent, the creation and annihilation operators are not, and they are responsible for creating and destroying particles in the quantum field theory. The **vacuum state** $|0\rangle$ is defined as the state that is annihilated by all annihilation operators:

$$\hat{a}_{\mathbf{k}} |0\rangle = 0 \quad \forall \mathbf{k},$$

while we can create one-particle states by acting with the creation operators on the vacuum, which adds one quanta of the field with momentum $\mathbf{k}$:

$$|\mathbf{k}\rangle = \sqrt{2\omega_{\mathbf{k}}} \hat{a}_{\mathbf{k}}^{\dagger} |0\rangle,$$

and more generally, we can create multi-particle states by acting with multiple creation operators on the vacuum, which leads to the **Fock space** of the theory. With this definition of single particle states, we can compute the inner product between two single particle states, which is given by

$$\langle \mathbf{k} | \mathbf{k}' \rangle = 2\omega_{\mathbf{k}} \delta^3(\mathbf{k} - \mathbf{k}'),$$

showing that the single particle states are orthogonal and normalized with respect to the relativistic normalization convention. The normalization $\sqrt{2\omega_{\mathbf{k}}}$ is arbitrary, but important for ensuring that the probabilities computed in the theory are consistent with relativistic invariance.

Remark. *To find a Lorentz invariant measure, we can start from the Minkowski space measure d^4k and impose the on-shell condition $k^2 = m^2$ using a delta function along with a step function to ensure that we only consider positive energy solutions:*

$$\int d^4k \delta(k^2 - m^2) \theta(k^0) = \int \frac{d^3\mathbf{k}}{2\omega_{\mathbf{k}}}.$$

We have then found the Lorentz invariant measure $\frac{d^3\mathbf{k}}{2\omega_{\mathbf{k}}}$ for integrating over the momentum space of on-shell particles, which is crucial for ensuring that the theory is consistent with special relativity. The previous normalization of the single particle states is consistent with this Lorentz invariant measure, since it ensures that the inner product between single particle states is Lorentz invariant, and it also allows us to compute probabilities and cross sections in a way that is consistent with relativistic invariance:

$$\int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| = 1,$$

which is the completeness relation for the single particle states in the theory. This ensures that we can express any state in the Fock space as a superposition of single particle states, and it also allows us to compute transition amplitudes and probabilities in a way that is consistent with relativistic invariance.

1.1.1 | Feynman's Propagator

1.1.2 | Noether's Theorem

1.2 | Spin 1/2

1.3 | Spin 1

2 | Interacting Fields

2.0.1 | Dimensional Analysis

2.1 | S-Matrix

Appendices

A | Notation and Conventions

Conventions on Units and Natural Units

We are going to use natural units throughout these notes, where the speed of light c and the reduced Planck constant $\hbar$ are set to 1:

$$c = 1, \quad \hbar = 1.$$

In natural units, all physical quantities are expressed in terms of a single unit of energy (or mass) and measured in Electronvolts (eV) and multiples thereof, since the dimensions of length and time are related to energy through the relations $E = mc^2$ and $E = \hbar\omega$. This simplifies many equations in quantum field theory and allows us to focus on the essential physics without being distracted by factors of c and $\hbar$. In these units, we have:

$$\begin{aligned} [E] &= [M] = [L]^{-1} = [T]^{-1}, \\ [\hbar] &= 1, \\ [c] &= 1. \end{aligned}$$

We report some useful conversion factors between natural units and SI units:

$$\begin{aligned} 1 \text{ GeV} &\approx 1.602 \times 10^{-10} \text{ J}, \\ 1 \text{ GeV} &\approx 1.783 \times 10^{-27} \text{ kg}, \\ 1 \text{ GeV}^{-1} &\approx 0.1975 \text{ fm}, \\ 1 \text{ GeV}^{-1} &\approx 6.582 \times 10^{-25} \text{ s}, \end{aligned}$$

and also

$$\begin{aligned} \hbar c &\approx 197.3 \text{ MeV} \cdot \text{fm}, \\ 1 \text{ fm} &\approx 10^{-15} \text{ m}, \end{aligned}$$

where the femtometer (fm) is a common unit of length in nuclear and particle physics, often called a Fermi: it is approximately the length of a proton or neutron radius:

$$r_p \sim 0.8 \text{ fm} = \frac{0.8}{197.3} \text{ MeV}^{-1}.$$

Conventions on Electromagnetism

In order to avoid confusion, we will use the **Heaviside-Lorentz** units for electromagnetism throughout these notes: in these units, the electric charge e is dimensionless and appears ex-

plicitly in the interaction terms of the Lagrangian, making it easier to identify the strength of electromagnetic interaction, while Maxwell equations take a more symmetric form.

Explicitly, in Heaviside-Lorentz units, we have the following relations:

- $\epsilon_0 = \mu_0 = 1$, where ϵ_0 is the *vacuum permittivity* and μ_0 is the *vacuum permeability*: this means that the electric and magnetic fields have the same units and that the speed of light c is equal to 1 (these units are consistent with the natural units we are using).
- The **electric charge** e is dimensionless and connected to the fine-structure constant α by

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{4\pi} \approx \frac{1}{137}.$$

- The **electric potential** Φ is defined as

$$\Phi = \frac{Q}{4\pi R},$$

where Q is the charge and R is the distance from the charge: this means that the electric potential has the same units as the electric field E (which is consistent with the fact that in Heaviside-Lorentz units, $\epsilon_0 = 1$).

- The **Maxwell equations** take the following form:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \rho, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t},\end{aligned}$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the charge density, and $\mathbf{J}$ is the current density: this symmetric form of Maxwell equations is one of the reasons why Heaviside-Lorentz units are often preferred in theoretical physics, especially in quantum field theory. Basically the Heaviside-Lorentz units are a rationalized version of the Gaussian units, where the factors of 4π are absorbed into the definitions of the fields and charges, leading to simpler equations and making it easier to identify the physical constants that govern the interactions.

Conventions on the Metric and Spacetime

We will use the **mostly minus** convention for the metric tensor $g_{\mu\nu}$ in Minkowski spacetime, which is defined as

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This means that the time component of the metric is positive, while the spatial components are negative; using Einstein summation convention, we have the following relations among the components of the metric and v^μ , w^μ four vectors:

$$\begin{aligned}v \cdot w &= g_{\mu\nu} v^\mu w^\nu = v^0 w^0 - \mathbf{v} \cdot \mathbf{w}, \\ v^2 &= v \cdot v = g_{\mu\nu} v^\mu v^\nu = (v^0)^2 - \mathbf{v}^2, \\ v_\mu &= g_{\mu\nu} v^\nu \implies v_0 = v^0, \quad \mathbf{v} = v^i = -v_i.\end{aligned}$$

We will use greek letters μ, ν, ρ, σ to denote spacetime indices that run from 0 to 3, while Latin letters i, j, k will be used to denote spatial indices that run from 1 to 3:

$$\begin{aligned}\mu, \nu, \rho, \sigma &\in \{0, 1, 2, 3\} = \{t, x, y, z\}, \\ i, j, k &\in \{1, 2, 3\} = \{x, y, z\}.\end{aligned}$$

Fourier Transforms

We will use the following conventions for Fourier transforms of a function $f(x)$ in four-dimensional spacetime:

$$\begin{aligned}f(x) &= \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot x} \tilde{f}(p), \\ \tilde{f}(p) &= \int d^4x e^{ip \cdot x} f(x).\end{aligned}$$

We highlight that the Fourier transform of a function $f(x)$ is denoted by $\tilde{f}(p)$, and that the integration measure includes a factor of $(2\pi)^{-4}$ in the inverse Fourier transform, which is a common convention in quantum field theory: in the space of coordinates, we have the measure d^4x , while in the momentum space, we have the measure $\frac{d^4p}{(2\pi)^4}$. This convention ensures that the Fourier transform and its inverse are consistent and that the normalization of the fields and operators in quantum field theory is correct.

Feynman Diagrams

In these notes, we will use Feynman diagrams as a graphical representation of the interactions between particles and fields in quantum field theory. We will use the following conventions for Feynman diagrams:

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- Time will be represented on the horizontal axis, going from left to right, which is a common convention in quantum field theory; this means that the left side of the diagram will represent the initial state of the particles, while the right side will represent the final state after the interaction has occurred.
- Solid lines will represent fermions (spin-1/2 particles), while dashed lines will represent bosons (spin-0 or spin-1 particles).
- Arrows on the lines will indicate the direction of particle flow, with arrows pointing towards the interaction vertex for incoming particles and away from the vertex for outgoing particles.
- Wavy lines will represent gauge bosons (spin-1 particles that mediate interactions), such as photons, W and Z bosons, and gluons.
- The vertices in the diagrams will represent interaction points where particles interact according to the rules of the theory, with each vertex corresponding to a specific term in the Lagrangian of the quantum field theory.

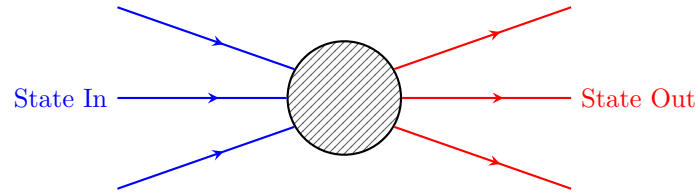


Figure 1: Generic Feynman diagram representing an interaction between particles, where the blue lines represent incoming particles (initial state) and the red lines represent outgoing particles (final state). The black box represents the interaction vertex, which corresponds to a specific term in the Lagrangian of the quantum field theory.

B | Computations and further developments